



(Don't Write anything in this Area)

- 6.c) How does 'Natural Selection' operate to bring about changes in gene frequencies? Illustrate with a human example. (15 Marks)

'Natural selection' or the theory of the survival of the fittest was given by Darwin in the 'Origin of Species'

Working of the 'Natural Selection' in gene frequency changes by:—

- ① Balanced Natural selection :→ Here the nature selects the one who has the stability to survive in the given environment.

Examples

- ① In the sickle-cell disease the heterozygotes are preferred in the environment where there is high chance of malaria. (SS)
- SS = normal = die because of Malaria
- ss = sickle cell death.

① Study by Sara Tiskoff the T6PD deficiency in Native Americans, here they had the advantage to survive in malaria condition.

② Case of Disruptive selection
Nature selects the genotype which are not better but they are more suitable to live in the environmental condition.

Examples

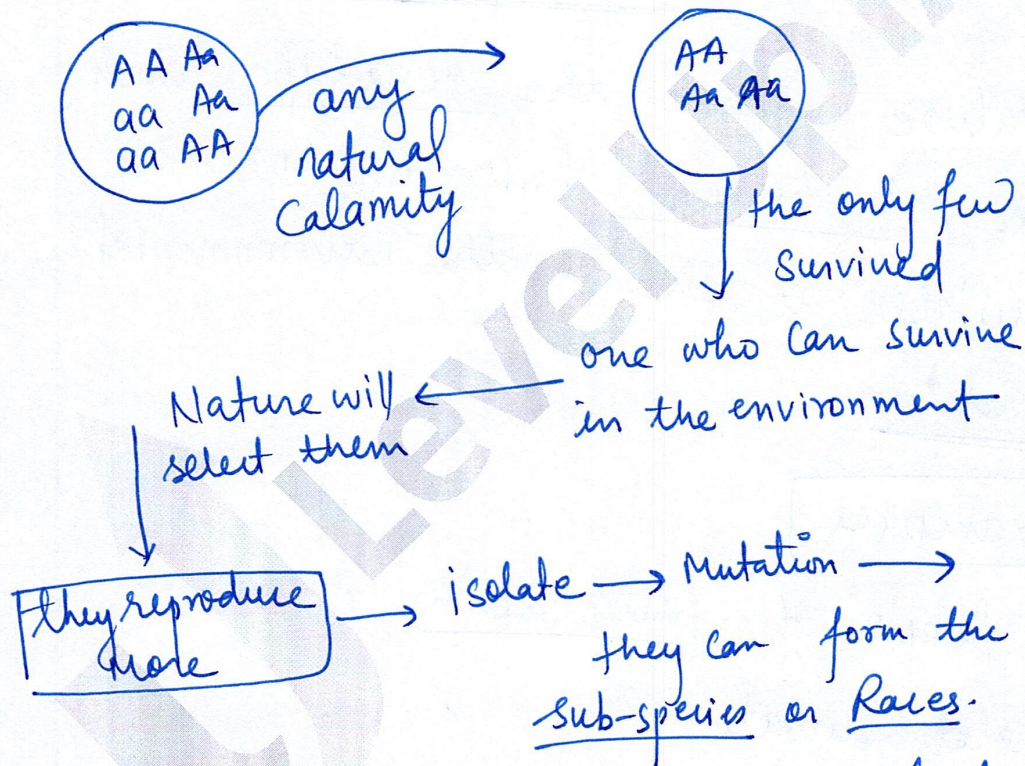
① People of fatty skin in the cold climate

② People of lean & thin body in the warm climate.

③ People having larger lung capacity or lesser metabolism in high altitude

③ Natural selection further works to select the male or female out of third gender also.

Explaining the process with example



(eg) In 1780, Pingalapese people had genetic drift, and they had only 30% carrier of color blind & 10% → Color blind.

Thus, we have seen how the natural selection had worked.